IE 622 Complete Solutions
(from `ps1_merged.pdf` and `Quiz1_merged.pdf`)

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Contents

1 Practice Problem Set 1 **3**

2 Practice Problem Set 2 **5**

3 Practice Problem Set 3 **6**

4 Practice Problem Set 4 **7**

5 Practice Problem Set 5 **10**

6 Practice Problem Set 6 **12**

7 Practice Problem Set 7 **13**

8 Practice Problem Set 8 **16**

9 Quiz 1 **17**

10 Quiz 2 **18**

11 Quiz 3 **18**

12 Quiz 4 **19**

13 Quiz 5 **19**

14 Quiz 6 **19**

15 Quiz 7 **20**

16 Practice Problem Set 9 **20**

17 Practice Problem Set 10 **21**

18 Practice Problem Set 11 **24**

19 Practice Problem Set 12 **26**

20 Practice Problem Set 13 **27**

21 Quiz 8 **28**

22 Quiz 9 **29**

23 Quiz 10 **29**

24 Quiz 11 **30**

25 Quiz 12 **30**

26 Quiz 13 **31**

1 Practice Problem Set 1

Q1

Because $\mathcal{F}$ is a σ-field, it is closed under complements and countable unions. Hence

$$\bigcap_{i\geq 1} A_i = \left(\bigcup_{i\geq 1} A_i^c\right)^c \in \mathcal{F}.$$

So $\cap_i A_i$ is an event.

Q2

Let $A_n \uparrow A$.

(a) $A = \lim_n A_n = \bigcup_{n\geq 1} A_n \in \mathcal{F}$ (closure under countable unions).

(b) Define $B_1 = A_1$, $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. Then (B_n) are disjoint,

$$A_n = \bigcup_{k=1}^{n} B_k, \qquad A = \bigcup_{k=1}^{\infty} B_k.$$

By countable additivity,

$$\mathbb{P}(A_n) = \sum_{k=1}^{n} \mathbb{P}(B_k) \uparrow \sum_{k=1}^{\infty} \mathbb{P}(B_k) = \mathbb{P}(A).$$

Hence $\mathbb{P}(A_n) \uparrow \mathbb{P}(A)$.

(c) If (E_k) are disjoint and $A_n = \cup_{k=1}^{n} E_k$, then $A_n \uparrow \cup_{k\geq 1} E_k$. Using part (b),

$$\mathbb{P}\left(\bigcup_{k\geq 1} E_k\right) = \lim_{n\to\infty} \mathbb{P}(A_n) = \lim_{n\to\infty} \sum_{k=1}^{n} \mathbb{P}(E_k) = \sum_{k\geq 1} \mathbb{P}(E_k),$$

which is countable additivity.

Q3

For any event A, the collection $\{\varnothing, \Omega, A, A^c\}$ contains $\varnothing, \Omega$, is closed under complement, and closed under countable unions (there are only finitely many sets to check). Hence it is a σ-field, i.e. $\sigma(A) = \{\varnothing, \Omega, A, A^c\}$.

Q4

Take $\Omega = \{1, 2, 3, 4\}$ with uniform probability and define

$$E_1 = \{1, 2\}, \quad E_2 = \{1, 3\}, \quad E_3 = \{1, 4\}.$$

Each has probability $1/2$, and each pairwise intersection has probability $1/4 = (1/2)(1/2)$, so they are pairwise independent. But

$$\mathbb{P}(E_1 \cap E_2 \cap E_3) = \mathbb{P}(\{1\}) = \frac{1}{4} \neq \frac{1}{8} = \prod_{i=1}^{3} \mathbb{P}(E_i),$$

so they are not mutually independent.

Q5

If $X \sim \text{Exp}(\lambda)$:

(a) $f_X(x) = \lambda e^{-\lambda x}\, \mathbf{1}_{\{x \geq 0\}}$.

(b) $\mathbb{P}(X > t) = e^{-\lambda t}$ for $t > 0$.

(c)
$$\mathbb{P}(X > t + s \mid X > s) = \frac{\mathbb{P}(X > t + s)}{\mathbb{P}(X > s)} = \frac{e^{-\lambda(t+s)}}{e^{-\lambda s}} = e^{-\lambda t}.$$

Q6

If F is continuous and $U = F(X)$, then for $u \in [0, 1]$,
$$\mathbb{P}(U \leq u) = \mathbb{P}(F(X) \leq u) = \mathbb{P}(X \leq F^{-1}(u)) = u.$$
So $U \sim \text{Unif}(0, 1)$.

Q7

(a) For $R = X/Y$ (assuming a continuous joint density):
$$f_R(r) = \int_{-\infty}^{\infty} |y|\, f_{X,Y}(ry, y)\, dy.$$
For the standard bivariate normal density
$$f_{X,Y}(x, y) = \frac{1}{2\pi} e^{-(x^2+y^2)/2},$$
this gives
$$f_R(r) = \frac{1}{\pi(1 + r^2)},$$
i.e. $R \sim \text{Cauchy}(0, 1)$.

(b) If (X, Y) is standard bivariate normal with independent components, then
$$X + Y \sim N(0, 2).$$

(c) Let $U = \min\{X, Y\}$ and $V = \max\{X, Y\}$. For $u \leq v$,
$$f_{U,V}(u, v) = f_{X,Y}(u, v) + f_{X,Y}(v, u),$$
and $f_{U,V}(u, v) = 0$ for $u > v$.

Q8

(a) If $X \sim \text{Poi}(\lambda)$ and $Y \sim \text{Poi}(\mu)$ are independent,
$$\mathbb{P}(X + Y = n) = \sum_{k=0}^{n} e^{-\lambda} \frac{\lambda^k}{k!}\, e^{-\mu} \frac{\mu^{n-k}}{(n-k)!} = e^{-(\lambda+\mu)} \frac{(\lambda + \mu)^n}{n!}.$$
So $X + Y \sim \text{Poi}(\lambda + \mu)$.

(b) If $X \sim \mathrm{Bin}(n,p)$ and $Y \sim \mathrm{Bin}(m,p)$ independent,

$$\mathbb{P}(X+Y=r) = \sum_k \binom{n}{k}\binom{m}{r-k} p^r (1-p)^{m+n-r} = \binom{m+n}{r} p^r (1-p)^{m+n-r},$$

so $X+Y \sim \mathrm{Bin}(m+n,p)$.

(c) For independent integer-valued X, Y with pmfs f, g,

$$\mathbb{P}(X+Y=n) = \sum_{k\in\mathbb{Z}} \mathbb{P}(X=k, Y=n-k) = \sum_{k\in\mathbb{Z}} f(k)g(n-k).$$

2 Practice Problem Set 2

Q1

(a) $\sigma(X) = \{X^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}$ is a σ-field since preimages preserve $\varnothing$, complements, and countable unions.

(b) If $X = \mathbf{1}_B$, then $\sigma(X) = \{\varnothing, \Omega, B, B^c\}$. If $X = \mathbf{1}_A + \mathbf{1}_B$ with $A \cap B = \varnothing$, then $X = \mathbf{1}_{A\cup B}$, so

$$\sigma(X) = \{\varnothing, \Omega, A\cup B, (A\cup B)^c\}.$$

(c)

$$\sigma(X_1, X_2) = \sigma\big((X_1, X_2)^{-1}(\mathcal{B}(\mathbb{R}^2))\big).$$

More generally,

$$\sigma(X_1, \ldots, X_m) = \sigma\big((X_1, \ldots, X_m)^{-1}(\mathcal{B}(\mathbb{R}^m))\big).$$

Q2

For $0 < m < n$, by Markov factorization,

$$\mathbb{P}(X_m = y \mid X_i = x_i,\ i \neq m) = \frac{p_{x_{m-1},y} p_{y,x_{m+1}}}{\sum_z p_{x_{m-1},z} p_{z,x_{m+1}}},$$

which depends only on (x_{m-1}, x_{m+1}). Hence

$$\mathbb{P}(X_m = y \mid X_i = x_i,\ i \neq m) = \mathbb{P}(X_m = y \mid X_{m-1} = x_{m-1}, X_{m+1} = x_{m+1}).$$

Q3

For $n_1 < \cdots < n_k$,

$$\mathbb{P}(X_{n_1} = i_{n_1}, \ldots, X_{n_k} = i_{n_k}) = \big(\lambda P^{n_1}\big)_{i_{n_1}} \prod_{r=2}^{k} p_{i_{n_{r-1}}, i_{n_r}}^{(n_r - n_{r-1})}.$$

Thus all such finite-dimensional distributions are defined from path probabilities.

Q4

Let $S_n = \sum_{m=1}^{n} X_m$, $Y_n = X_n + X_{n-1}$, $Z_n = \sum_{m=1}^{n} S_m$.

(a) (S_n) is a DTMC (random walk with i.i.d. increments).

(b) (S_{2n}) is also a DTMC (increments are i.i.d. $X_{2n+1} + X_{2n+2}$).

(c) (Y_n) is generally *not* Markov (knowledge of Y_n does not determine distribution of Y_{n+1}).

(d) (Z_n) is generally *not* Markov (next step depends on S_{n+1}, not determined by Z_n alone).

(e) $((S_n, Z_n))$ *is* a DTMC, since

$$S_{n+1} = S_n + X_{n+1}, \qquad Z_{n+1} = Z_n + S_{n+1},$$

and X_{n+1} is independent of past.

Q5

A first-order Markov model is obtained by the 2-bit state

$$W_n = (Y_{n-1}, Y_n) \in \{00, 01, 10, 11\}, \ n \geq 1.$$

Transitions:

- $00 \to 01$ with prob. 1,
- $11 \to 10$ with prob. 1,
- $01 \to 10$ with prob. $1-p$, and $01 \to 11$ with prob. p,
- $10 \to 00$ with prob. $1-p$, and $10 \to 01$ with prob. p.

With order $(00, 01, 10, 11)$,

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1-p & p \\ 1-p & p & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Q6 (fill in)

One valid line: "Past and future talk to each other only through the present."

3 Practice Problem Set 3

Q1

For $H^A = \inf\{n \geq 0 : X_n \in A\}$,

$$\{H^A \leq n\} = \bigcup_{k=0}^{n} \{X_k \in A\} \in \mathcal{F}.$$

Hence H^A is measurable, so it is a random variable.

Q2

(a) $\min\{S, T\}$ is a stopping time since $\{\min(S, T) \le n\} = \{S \le n\} \cup \{T \le n\}$.

(b) $S - 1$ is generally *not* a stopping time (depends on one-step-ahead information).

(c) $T + 1$ is a stopping time since $\{T + 1 \le n\} = \{T \le n - 1\} \in \mathcal{F}_n$.

(d) "Last visit time" is *not* a stopping time (needs future knowledge).

(e) A constant k is a stopping time.

Q3

(a) Each T_m is a stopping time (proved recursively from stopping-time closure and adaptedness of X_n).

(b) $Y_m := X_{T_m}$ is a DTMC by the strong Markov property at T_m. Its state space is the subset of A that can be entered from A^c (typically all of A in irreducible finite chains).

Q4

(a) Classes are $\{1, 2, 5\}$ and $\{3, 4\}$. Closed class: $\{3, 4\}$. Not irreducible.

(b) For the 6×6 matrix: classes are $\{1, 2, 3\}$, $\{4\}$, and $\{5, 6\}$. Closed class: $\{5, 6\}$ only. Not irreducible.

Q5

(a) If C is closed ($p_{ij} = 0$ for $i \in C, j \notin C$), then any path starting in C cannot leave C, so $i \to j$ with $i \in C$ implies $j \in C$. Conversely, if this implication holds and some $p_{ij} > 0$ with $i \in C, j \notin C$, then $i \to j$ contradicts the implication.

(b) Finite state-space: communication classes form a finite DAG; every finite DAG has a sink class, which is closed. Example with no closed class (infinite state-space): $p_{i,i+1} = 1$ on $\mathbb{N}$.

Q6 (fill in)

One valid line: "Use the present wisely, and even stopping times become milestones."

4 Practice Problem Set 4

Q1 (Gambler's ruin)

Let $p = 0.4$, $q = 0.6$, start $i = 3$, target $N = 8$.

(a) Timid strategy (± 1 walk):

$$\mathbb{P}_i(\tau_N < \tau_0) = \frac{1 - (q/p)^i}{1 - (q/p)^N} = \frac{1 - 1.5^3}{1 - 1.5^8} \approx 0.09643.$$

(b) Bold strategy:

$$B_4 = 0.4, \quad B_6 = 0.4 + 0.6B_4 = 0.64, \quad B_3 = 0.4B_6 = 0.256.$$

(c) Bold is better: $0.256 > 0.09643$.

Q2 (Pattern waiting times)

Using standard automaton-state equations:

$$\mathbb{E}[T_{\text{HTH}}] = 10, \qquad \mathbb{E}[T_{\text{HHH}}] = 14, \qquad \mathbb{E}[T_{\text{THTH}}] = 20.$$

Q3

Reading assignment item; no derivation requested.

Q4

If i is recurrent and $f_{ij} > 0$, then from repeated returns to i and Markov property, $f_{ij} = 1$. If $f_{ji} < 1$, then starting at i there is positive probability to hit j and then never return to i, contradicting recurrence of i. Hence $f_{ji} = 1$. Therefore

$$f_{jj} \geq f_{ji} f_{ij} = 1,$$

so j is recurrent.

Q5

Let $N_i = \sum_{n \geq 0} \mathbf{1}_{\{X_n = i\}}$. Then

$$\{X_n = i \text{ i.o.}\} = \{N_i = \infty\}.$$

State i is recurrent iff $\mathbb{P}_i(N_i = \infty) = 1$, and transient iff $\mathbb{P}_i(N_i < \infty) = 1$, equivalently

$$\text{transient} \iff \mathbb{P}_i(X_n = i \text{ i.o.}) = 0.$$

Q6

Let $T_j = \inf\{n \geq 1 : X_n = j\}$ and $N_j = \sum_{n \geq 1} \mathbf{1}_{\{X_n = j\}}$. Condition on whether j is ever hit:

$$\mathbb{E}_i[N_j] = f_{ij} \cdot \mathbb{E}_j[N_j \text{ after a visit}] = f_{ij} \cdot \frac{1}{1 - f_{jj}}.$$

Hence

$$\mathbb{E}_i[N_j] = \frac{f_{ij}}{1 - f_{jj}}.$$

Q7

(a) For matrix (a), recurrent states: $\{3, 4\}$ (closed class). Transient: $\{1, 2, 5\}$. Stationary distribution:

$$\pi = (0, 0, 4/5, 1/5, 0).$$

(b) For matrix (b), closed class is $\{5, 6\}$, so $5, 6$ recurrent and $1, 2, 3, 4$ transient. Stationary distribution:

$$\pi = (0, 0, 0, 0, 1/2, 1/2).$$

Q8 (Umbrellas)

With N umbrellas total, stationary distribution for umbrellas at departure location is

$$\pi_0 = \frac{1-p}{N+1-p}, \qquad \pi_i = \frac{1}{N+1-p}, \ i = 1, \ldots, N.$$

You get wet iff it rains and there are 0 umbrellas at departure location, so

$$\mathbb{P}(\text{wet}) = p\pi_0 = \frac{p(1-p)}{N+1-p}.$$

For the given $N = 4$:

$$\mathbb{P}(\text{wet}) = \frac{p(1-p)}{5-p}.$$

If $p = 0.6$, this is $0.24/4.4 \approx 0.0545$. To make wet probability < 0.1 when $p = 0.6$:

$$\frac{0.24}{N+0.4} < 0.1 \iff N > 2,$$

so minimum is $N = 3$ umbrellas.

Q9 (Knight walk)

(a) Markov: next move depends only on current square.

(b) State space: 64 squares.

(c) Irreducible: yes (knight graph on 8×8 board is connected).

(d) Stationary distribution:

$$\pi(v) = \frac{\deg(v)}{\sum_u \deg(u)} = \frac{\deg(v)}{336}.$$

Interpretation: long-run fraction of time on square v.

(e) Most likely squares: degree 8 (center region), each with probability $1/42$. Least likely: corners (degree 2), each with probability $1/168$.

Q10

For increments $+2$ (prob. p) and -1 (prob. $1-p$), mean drift is

$$\mu = 2p - (1-p) = 3p - 1.$$

The walk is recurrent only at zero drift ($p = 1/3$). For $p \neq 1/3$, it is transient.

Q11

Example:

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1/2 & 1/2 & 0 \end{pmatrix}.$$

Then every

$$\pi = (\alpha, 1-\alpha, 0), \quad \alpha \in [0,1],$$

is stationary. So non-irreducible chains can have infinitely many stationary distributions.

Q12 (fill in)

One valid line: "I move by local rules, but in the long run the board writes the law."

5 Practice Problem Set 5

Q1 (Elevator chain)

From state 0, jump to height $K \geq 1$ with $\mathbb{P}(K = k) = q_k$; then descend deterministically to 0.

(a) Recurrent: return to 0 occurs almost surely in finite time each cycle.

(b) Mean return time:

$$\mathbb{E}_0[T_0^+] = 1 + \mathbb{E}[K] = 1 + \sum_{k \geq 1} k q_k.$$

So positive recurrence iff $\sum_{k \geq 1} k q_k < \infty$.

Q2 (Limit of P^n)

Closed classes: $\{1\}$ and $\{2, 4\}$ (with stationary distribution on $\{2, 4\}$ equal to $(1/3, 2/3)$). States $3, 5$ are transient. Absorption probabilities into state 1:

$$a_1 = 1,\ a_2 = 0,\ a_3 = 1/3,\ a_4 = 0,\ a_5 = 2/3.$$

Hence

$$\lim_{n \to \infty} p_{i1}^{(n)} = a_i, \quad \lim_{n \to \infty} p_{i2}^{(n)} = \frac{1 - a_i}{3}, \quad \lim_{n \to \infty} p_{i4}^{(n)} = \frac{2(1 - a_i)}{3},$$

and limits to states $3, 5$ are 0. Rows:

$$\begin{aligned}
i = 1 &: (1, 0, 0, 0, 0), \\
i = 2 &: (0, 1/3, 0, 2/3, 0), \\
i = 3 &: (1/3, 2/9, 0, 4/9, 0), \\
i = 4 &: (0, 1/3, 0, 2/3, 0), \\
i = 5 &: (2/3, 1/9, 0, 2/9, 0).
\end{aligned}$$

Q3

Given

$$P = \begin{pmatrix} r & 1 - r & 0 \\ q & 0 & p \\ 0 & s & 1 - s \end{pmatrix}, \quad p + q = 1,$$

start from 0, and $N_j = \sum_{k=0}^{T_0 - 1} \mathbf{1}_{\{X_k = j\}}$.

(a)

$$\gamma_{00} = \mathbb{E}_0[N_0] = 1, \qquad \gamma_{10} = \frac{1 - r}{q}, \qquad \gamma_{20} = \frac{p(1 - r)}{sq}.$$

(b) Stationary distribution:

$$\pi_1 = \frac{1 - r}{q} \pi_0, \qquad \pi_2 = \frac{p(1 - r)}{sq} \pi_0,$$

$$\pi_0 = \left(1 + \frac{1-r}{q} + \frac{p(1-r)}{sq}\right)^{-1}.$$

So indeed

$$\pi_j = \frac{\gamma_{j0}}{\gamma_{00} + \gamma_{10} + \gamma_{20}}, \quad j = 0, 1, 2.$$

Q4 (Durrett 1.74)

For chain with $p_{i,i+1} = q_i$ and $p_{i,0} = 1 - q_i$, define

$$a_n := \prod_{k=0}^{n-1} q_k, \quad n \geq 1, \qquad a_0 := 1.$$

Then

$$\mathbb{P}_0(T_0 > n) = a_n,$$

so recurrence is equivalent to

$$\mathbb{P}_0(T_0 < \infty) = 1 \iff \lim_{n\to\infty} a_n = 0 \iff \prod_{k=0}^{\infty} q_k = 0.$$

Mean return time:

$$\mathbb{E}_0[T_0] = \sum_{n\geq 0} \mathbb{P}_0(T_0 > n) = \sum_{n\geq 0} a_n.$$

Hence positive recurrence iff $\sum_{n\geq 0} a_n < \infty$. When positive recurrent,

$$\pi_i = \pi_0 a_i, \qquad \pi_0 = \left(\sum_{n\geq 0} a_n\right)^{-1}.$$

Q5

If $i \leftrightarrow j$, choose m, n with $p_{ij}^{(m)} > 0$, $p_{ji}^{(n)} > 0$. For any t with $p_{ii}^{(t)} > 0$, we have $p_{jj}^{(m+t+n)} > 0$. Thus $d(j)$ divides all numbers $m + t + n$, so $d(j) \mid d(i)$. Symmetrically $d(i) \mid d(j)$. Hence $d(i) = d(j)$.

Q6

(a) For the first matrix (6 states), all states communicate. Return cycles of lengths 3 (via $1 \to 2 \to 3 \to 1$) and 4 (via $3 \to 4 \to 5 \to 6 \to 3$) exist; $\gcd(3, 4) = 1$. So all states have period 1 (aperiodic).

(b) For the second matrix, classes are $\{1, 2, 3\}$ and $\{4, 5\}$. In $\{1, 2, 3\}$, return lengths include 3 and 5, so period 1. In $\{4, 5\}$ both states have self-loops, so period 1. Hence all states are aperiodic.

6 Practice Problem Set 6

Q1

Given

$$X_{n+1} = (X_n + Y_{n+1})^+, \quad \mathbb{P}(Y_n = 2) = p,\ \mathbb{P}(Y_n = -1) = 1 - p.$$

(a) DTMC on $\{0, 1, 2, \dots\}$ with transitions

$$0 \to 0\ (1-p),\ 0 \to 2\ (p), \qquad i \geq 1:\ i \to i-1\ (1-p),\ i \to i+2\ (p).$$

(b) Transition diagram follows directly from the above arrows.

(c) With $V(x) = x$, for $i \geq 1$,

$$\Delta V(i) = \mathbb{E}[X_{n+1} - X_n \mid X_n = i] = 2p - (1-p) = 3p - 1.$$

So a sufficient condition for positive recurrence is $p < 1/3$ (negative drift outside a finite set).

(d) For $p < 1/3$ (positive recurrent), use ergodic theorem/renewal reward:

$$\frac{1}{n}\mathbb{E}[Z_n] \to \pi_0 = \frac{1}{\mathbb{E}_0[T_0^+]},$$

where π_0 is stationary mass at 0.

Q2

Here $p_{0i} = \alpha/i^2$ $(i \geq 1)$ with $\sum_{i\geq 1} \alpha/i^2 = 1$, and for $i \geq 1$, $p_{i,i+1} = p$, $p_{i,i-1} = q = 1 - p$, $p < 1/2$.

(a) Diagram: from 0 jump to $i \geq 1$ with probability α/i^2; from each $i \geq 1$, move to $i+1$ with p and to $i-1$ with q.

(b) With $V(i) = i$, for $i \geq 1$,

$$\Delta V(i) = p - q = -(q - p) < 0.$$

Foster recurrence criterion gives $f_{00} = 1$.

(c) There exists $\varepsilon = q - p > 0$ such that

$$\mathbb{E}[X_{n+1} - X_n \mid X_n = i] = -(q-p) \leq -\varepsilon, \quad i \geq 1.$$

Yet the chain is null recurrent: from $i \geq 1$, $\mathbb{E}_i[T_0] = i/(q-p)$ (solve one-step recursion), so

$$\mathbb{E}_0[T_0^+] = 1 + \sum_{i\geq 1} \frac{\alpha}{i^2} \cdot \frac{i}{q-p} = 1 + \frac{\alpha}{q-p} \sum_{i\geq 1} \frac{1}{i} = \infty.$$

Thus recurrent but not positive recurrent.

(d) No violation. Positive-recurrence Lyapunov theorem requires additional hypotheses (notably finite $\mathbb{E}[V(X_{n+1}) \mid X_n = x]$ on a finite set). Here at $x = 0$, with $V(i) = i$, that expectation is infinite due harmonic divergence.

Q3 (Reversibility)

(a)

$$P = \begin{pmatrix} 0 & p & 1-p \\ 1-p & 0 & p \\ p & 1-p & 0 \end{pmatrix}$$

is reversible iff Kolmogorov cycle products match:

$$p^3 = (1-p)^3 \iff p = 1/2.$$

(b) Any finite nearest-neighbor (birth-death) chain on $\{0, 1, \ldots, N\}$ is reversible (construct π recursively by detailed balance).

(c) If $p_{ij} = p_{ji}$ for all i, j, then chain is reversible with respect to uniform π (finite state-space).

(d) For $p_{01} = 1$, $p_{i,i+1} = p$, $p_{i,i-1} = 1-p$ $(i \geq 1)$, detailed balance gives

$$\pi_{i+1} = \frac{p}{1-p}\pi_i, \ i \geq 1, \qquad \pi_1 = \frac{1}{1-p}\pi_0.$$

A normalizable stationary distribution exists iff $p < 1/2$. Hence reversible (as a probability chain) iff $p < 1/2$.

Q4

(a) For simple random walk on connected undirected graph,

$$\pi_i = \frac{d_i}{2|E|}, \qquad P_{ij} = \frac{1}{d_i} \text{ if } (i,j) \in E,$$

so

$$\pi_i P_{ij} = \frac{1}{2|E|} = \pi_j P_{ji}$$

on edges, and both sides are 0 off edges. So detailed balance holds.

(b) For knight walk on chessboard,

$$\pi(v) = \frac{\deg(v)}{336}.$$

Hence per-square probabilities by degree are: $2/336, 3/336, 4/336, 6/336, 8/336$.

Q5 (fill in)

One valid line: "Reverse me in time and I still obey detailed balance."

7 Practice Problem Set 7

Q1

(a) If $\mathbb{P}(A_n) = 1$ for all n, then

$$\mathbb{P}\Big(\bigcap_n A_n\Big) = 1 - \mathbb{P}\Big(\bigcup_n A_n^c\Big) \geq 1 - \sum_n \mathbb{P}(A_n^c) = 1.$$

Also $\mathbb{P}(\cup_n A_n) = 1$.

(b) If $\mathbb{P}(B) = 1$, then

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \mathbb{P}(A \cap B) = \mathbb{P}(A)$$

since $\mathbb{P}(B^c) = 0$.

(c) If $\mathbb{E}|X| < \infty$, then $\mathbb{P}(|X| = \infty) = 0$; otherwise $|X|$ would be infinite on a set of positive probability and expectation would be infinite.

Q2

(a)

$$\limsup A_n = \bigcap_{m \geq 1} \bigcup_{n \geq m} A_n, \qquad \liminf A_n = \bigcup_{m \geq 1} \bigcap_{n \geq m} A_n,$$

so both are in $\mathcal{F}$.

(b) Immediate from definitions: $\liminf A_n \subseteq \limsup A_n$.

(c)

$$\{\lim X_n = X\} = \bigcap_{k \geq 1} \bigcup_{N \geq 1} \bigcap_{n \geq N} \{|X_n - X| < 1/k\} \in \mathcal{F}.$$

(d) For each ω, $\limsup \mathbf{1}_{A_n}(\omega) = 1$ iff $\omega \in A_n$ infinitely often, i.e. iff $\omega \in \limsup A_n$. Hence

$$\mathbf{1}_{\limsup A_n}(\omega) = \limsup_n \mathbf{1}_{A_n}(\omega).$$

Q3 (Infinite monkeys)

Let E_m be the event that positions $6m + 1$ to $6m + 6$ spell BANANA. Then

$$\mathbb{P}(E_m) = 26^{-6} > 0,$$

and (E_m) are independent (disjoint blocks). Since

$$\sum_m \mathbb{P}(E_m) = \infty,$$

Borel–Cantelli (II) gives $\mathbb{P}(E_m \text{ i.o.}) = 1$. Therefore BANANA appears infinitely often almost surely.

Q4 (Online dating)

Assume independent daily picks and deterministic member counts. Let M_n be male count, F_n female count on day n.

(a) If Bob only dates when Alice picks him,

$$\mathbb{P}(\text{meet on day } n) = \frac{1}{M_n}, \quad M_n = M_0 + 50n.$$

Since $\sum_n 1/M_n = \infty$ and events are independent across days, they meet infinitely often a.s.; they do *not* eventually stop meeting.

(b) If both must pick each other,

$$\mathbb{P}(\text{meet on day } n) = \frac{1}{M_n F_n}, \quad F_n = F_0 + 40n.$$

Now $\sum_n 1/(M_n F_n) < \infty$ (order $1/n^2$), so meetings occur only finitely many times a.s.; they eventually stop meeting.

(c) If male count grows by 1% daily, $M_n \approx M_0(1.01)^n$, then

$$\sum_n \frac{1}{M_n} < \infty,$$

so meetings are finite a.s.; they eventually stop meeting.

Q5

Counterexample on a common space: let $Y \sim \text{Bernoulli}(1/2)$ and define $Y_n = 1 - Y$ for all n. Then $Y_n \overset{d}{=} Y$ for every n, so $Y_n \xrightarrow{d} Y$. But

$$\mathbb{P}(|Y_n - Y| > 0) = 1,$$

so $Y_n \overset{\mathbb{P}}{\not\to} Y$.

Q6

(a) If X_n are i.i.d. with $\mathbb{E}|X_1| < \infty$ and $Y_n = \mathbf{1}_{\{|X_n| \le n\}} X_n$, then

$$\mathbb{E}[Y_n] = \mathbb{E}[X_1 \mathbf{1}_{\{|X_1| \le n\}}] \to \mathbb{E}[X_1]$$

by dominated convergence.

(b) Not true without i.i.d. Counterexample:

$$X_n = \begin{cases} n^2, & \text{with prob. } 1/n, \\ -n, & \text{with prob. } 1 - 1/n. \end{cases}$$

Then $\mathbb{E}[X_n] = 1 = \mathbb{E}[X_1]$ for all n, but

$$Y_n = \mathbf{1}_{\{|X_n| \le n\}} X_n = \begin{cases} 0, & X_n = n^2, \\ -n, & X_n = -n, \end{cases}$$

so $\mathbb{E}[Y_n] = -(n-1) \not\to 1$.

Q7 (fill in)

One valid line: "If your typing chance stays positive, I appear infinitely often."

8 Practice Problem Set 8

Q1

For right-continuous paths on discrete state space,

$$\{X_t = i \ \forall t \in [t_1, t_2]\} = \bigcap_{q \in \mathbb{Q} \cap [t_1, t_2]} \{X_q = i\}.$$

Hence

$$\mathbb{P}(X_t = i \ \forall t \in [t_1, t_2]) = \mathbb{P}\Big(\bigcap_{q \in \mathbb{Q} \cap [t_1, t_2]} \{X_q = i\} \Big),$$

which is the limit of probabilities of finite rational-grid intersections (finite-dimensional distributions).

Q2 (Exponential properties)

Let $S_i \sim \mathrm{Exp}(\lambda_i)$ independent.

(a) For $t \geq 0$,

$$\mathbb{P}(\lambda_1 S_1 > t) = \mathbb{P}(S_1 > t/\lambda_1) = e^{-t},$$

so $\lambda_1 S_1 \sim \mathrm{Exp}(1)$.

(b)

$$\mathbb{P}\Big(\min_{1 \leq i \leq n} S_i > t \Big) = \prod_{i=1}^{n} e^{-\lambda_i t} = e^{-(\lambda_1 + \cdots + \lambda_n)t}.$$

(c)

$$\mathbb{P}\Big(S_1 < \min_{2 \leq i \leq n} S_i \mid \min_i S_i > t \Big) = \frac{\lambda_1}{\lambda_1 + \cdots + \lambda_n}.$$

So this event is independent of $\{\min_i S_i > t\}$.

(d) If $\sum_n \lambda_n^{-1} < \infty$, then

$$\mathbb{E}\Big[\sum_n S_n \Big] = \sum_n \mathbb{E}[S_n] = \sum_n \frac{1}{\lambda_n} < \infty,$$

thus $\sum_n S_n < \infty$ a.s.

(e) If $\sum_n \lambda_n^{-1} = \infty$, for any $u > 0$,

$$\mathbb{E}\left[e^{-u \sum_{n=1}^{N} S_n} \right] = \prod_{n=1}^{N} \frac{\lambda_n}{\lambda_n + u}.$$

As $N \to \infty$, product tends to 0 (since $\sum_n \log(1 + u/\lambda_n) = \infty$), so

$$\mathbb{E}\left[e^{-u \sum_n S_n} \right] = 0.$$

Hence $\sum_n S_n = \infty$ a.s.

Q3

For Poisson process (X_t) of rate λ,

$$p_0(t+h) = \mathbb{P}(X_{t+h} = 0) = \mathbb{P}(X_t = 0)\mathbb{P}(X_{t+h} - X_t = 0) = p_0(t)e^{-\lambda h}.$$

So

$$\frac{p_0(t+h) - p_0(t)}{h} = p_0(t)\frac{e^{-\lambda h} - 1}{h} \to -\lambda p_0(t),$$

thus $p_0'(t) = -\lambda p_0(t)$.

Q4

For $0 \le t_1 < \cdots < t_n$ and integers $0 \le x_1 \le \cdots \le x_n$,

$$\mathbb{P}(X_{t_1} = x_1, \ldots, X_{t_n} = x_n) = \frac{e^{-\lambda t_1}(\lambda t_1)^{x_1}}{x_1!} \prod_{k=2}^{n} \frac{e^{-\lambda(t_k - t_{k-1})}\big(\lambda(t_k - t_{k-1})\big)^{x_k - x_{k-1}}}{(x_k - x_{k-1})!}.$$

(If $x_1 \le \cdots \le x_n$ fails, probability is 0.) These formulas determine all fdds.

Q5

If N_t is Poisson with rate 2:

(a)

$$\mathbb{P}(N_3 = 4 \mid N_1 = 1) = \mathbb{P}(N_3 - N_1 = 3) = e^{-4}\frac{4^3}{3!}.$$

(b) Given $N_3 = 4$, $N_1 \sim \mathrm{Bin}(4, 1/3)$, so

$$\mathbb{P}(N_1 = 1 \mid N_3 = 4) = \binom{4}{1}\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^3 = \frac{32}{81}.$$

Q6

Assume $t \ge s$. Then $N_t = N_s + (N_t - N_s)$ with independent increment $N_t - N_s$ independent of N_s. Hence

$$\mathrm{Cov}(N_t, N_s) = \mathrm{Cov}(N_s, N_s) = \mathrm{Var}(N_s) = \lambda s = \lambda \min\{t, s\}.$$

9 Quiz 1

Q1–Q5 (True/False with justification)

1. **True.** For decreasing A_n, continuity from above gives

$$\mathbb{P}(A_n) \downarrow \mathbb{P}\Big(\bigcap_{n \ge 1} A_n\Big).$$

2. **False in general.** The convolution pmf formula needs independence of X and Y.

3. **False.** $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ does not imply independence in general.

4. **False.** For exponential,
$$\mathbb{P}(X > t + s \mid X > s) = \mathbb{P}(X > t),$$
not strictly smaller.

5. **True.** For $Y = 1 - e^{-X}$ with $X \sim \text{Exp}(1)$,
$$\mathbb{P}(Y \le y) = \mathbb{P}(X \le -\log(1-y)) = y, \quad y \in [0,1].$$
So $Y \sim \text{Unif}(0,1)$.

10 Quiz 2

Use state $S_n = (Y_{n-2}, Y_{n-1}, Y_n) \in \{0,1\}^3$, $n \ge 2$. Define
$$r := \mathbb{P}(Y_{n+1} = 1 \mid Y_n = Y_{n-1} \ne Y_{n-2}) = \mathbb{P}(B \oplus X_{n+1} = 1) = 2p(1-p),$$
where $B \sim \text{Bernoulli}(p)$ is the extra flip coin in the rule. Then transitions are:

state abc	next-state probabilities for $(bc\,d)$
000	001 (1)
001	010 $(1-p)$, 011 (p)
010	100 $(1-p)$, 101 (p)
011	110 $(1-r)$, 111 (r)
100	000 $(1-r)$, 001 (r)
101	010 $(1-p)$, 011 (p)
110	100 $(1-p)$, 101 (p)
111	110 (1)

Initial distribution: $S_2 = (X_0, X_1, X_2)$ with i.i.d. Bernoulli(p) bits.

11 Quiz 3

For the given 5-state matrix:

1. Communicating classes are $\{1,2,5\}$ and $\{3,4\}$. A closed class is $C = \{3,4\}$.
2. Hitting probabilities of C:
$$h_i^C = 1 \text{ for all } i \in \{1,2,3,4,5\}.$$
3. With $D = I \setminus C = \{1,2,5\}$,
$$h_i^D = 1 \ (i \in D), \qquad h_3^D = h_4^D = 0.$$
4. For $i, j \in D$, $i \ne j$:
$$h_1^{\{2\}} = 0.9, \quad h_1^{\{5\}} = 1,$$
$$h_2^{\{1\}} = 1, \quad h_2^{\{5\}} = 1,$$
$$h_5^{\{1\}} = 0.1, \quad h_5^{\{2\}} = 0.1.$$

12 Quiz 4

Let $N_{j\to k}$ be the total number of $j \to k$ transitions, start from $X_0 = j$.

$$\mathbb{E}_j[N_{j\to k}] = \sum_{n\geq 0} \mathbb{P}_j(X_n = j, X_{n+1} = k) = \sum_{n\geq 0} \mathbb{P}_j(X_n = j)p_{jk} = p_{jk} \sum_{n\geq 0} \mathbb{P}_j(X_n = j).$$

Using $\sum_{n\geq 0} \mathbb{P}_j(X_n = j) = 1/(1 - f_{jj})$,

$$\mathbb{E}_j[N_{j\to k}] = \frac{p_{jk}}{1 - f_{jj}}.$$

13 Quiz 5

Chain: from $i \geq 1$, move to $i - 1$ deterministically; from 0, jump to $k \geq 0$ with probability q_k.

1. Recurrent: from any state, state 0 is hit in finite time almost surely, and this regenerates forever.

2. First return time from 0 is $T_0^+ = K + 1$ where $\mathbb{P}(K = k) = q_k$. So

$$\mathbb{E}_0[T_0^+] = 1 + \sum_{k\geq 1} kq_k.$$

   Positive recurrence iff $\sum_{k\geq 1} kq_k < \infty$.

3. When positive recurrent,

$$\pi_0 = \frac{1}{\mathbb{E}_0[T_0^+]} = \frac{1}{1 + \sum_{k\geq 1} kq_k}.$$

4. Since $q_0 > 0$, $p_{00} = q_0 > 0$, so period of 0 is 1. By irreducibility (all $q_k > 0$), all states are aperiodic.

5. If $q_{2m} = 0$ for all m (only odd jumps), then return times to 0 are $k + 1$ with odd k, hence all even. So $d(0) = 2$. For state 10, any return length is also even, hence $d(10) = 2$.

14 Quiz 6

For $i \geq 1$: $p_{i,i+2} = p$, $p_{i,i-1} = 1 - p$; and at 0: $p_{0,2} = q$, $p_{0,0} = 1 - q$.

Using $V(x) = x$:

$$\Delta V(i) = \mathbb{E}[X_{n+1} - X_n \mid X_n = i] = 3p - 1, \quad i \geq 1.$$

So a sufficient condition for positive recurrence is

$$p < 1/3.$$

Using $V'(x) = x^2$:

$$\Delta V'(i) = p\big((i + 2)^2 - i^2\big) + (1 - p)\big((i - 1)^2 - i^2\big) = (6p - 2)i + (3p + 1).$$

This is negative for all sufficiently large i iff $p < 1/3$. So V' gives the same threshold (no improvement in parameter condition).

15 Quiz 7

1. $$\liminf A_n = \bigcup_{m\geq 1} \bigcap_{n\geq m} A_n \in \mathcal{F}.$$

2. For every ω,
$$\mathbf{1}_{\liminf A_n}(\omega) = \liminf_n \mathbf{1}_{A_n}(\omega).$$

3. By Fatou on indicators,
$$\mathbb{P}(\liminf A_n) = \mathbb{E}[\mathbf{1}_{\liminf A_n}] = \mathbb{E}[\liminf \mathbf{1}_{A_n}] \leq \liminf \mathbb{E}[\mathbf{1}_{A_n}] = \liminf \mathbb{P}(A_n).$$

16 Practice Problem Set 9

Q1

Reading assignment: Theorem 2.4.6 in Norris. (No derivation required.)

Q2 (Splitting of a Poisson process)

Let $\Lambda = \lambda_1 + \lambda_2$ where $\lambda_1 = p\lambda$, $\lambda_2 = (1-p)\lambda$. For any non-negative integers m, n, independence of (θ_k) from (X_t) gives

$$\mathbb{P}(X_t^1 = m,\, X_t^2 = n) = \mathbb{P}(X_t = m{+}n)\binom{m+n}{m}p^m(1-p)^n = \frac{e^{-\lambda t}(\lambda t)^{m+n}}{(m+n)!} \cdot \frac{(m+n)!}{m!\,n!}p^m(1-p)^n$$
$$= \frac{e^{-p\lambda t}(p\lambda t)^m}{m!} \cdot \frac{e^{-(1-p)\lambda t}((1-p)\lambda t)^n}{n!}.$$

This factors as $\mathbb{P}(\mathrm{Poi}(p\lambda t) = m) \cdot \mathbb{P}(\mathrm{Poi}((1-p)\lambda t) = n)$, so X_t^1 and X_t^2 are independent for each t. For the full independence of the two processes, note that joint finite-dimensional distributions of $(X_{t_1}^1, \ldots, X_{t_k}^1, X_{s_1}^2, \ldots, X_{s_\ell}^2)$ factor because the labeling θ_n is i.i.d. and independent of X: conditioning on X and the inter-arrival structure and applying the same Binomial-splitting calculation to each increment interval yields the factored product form.

Q3 (First jump time from the infinitesimal definition)

Let $p_0(t) = \mathbb{P}(J_1 > t) = \mathbb{P}(X_t = 0)$. By independent increments and the infinitesimal characterisation,
$$p_0(t+h) = \mathbb{P}(X_t = 0)\mathbb{P}(X_{t+h} - X_t = 0) = p_0(t)(1 - \lambda h + o(h)).$$
Rearranging: $p_0'(t) = -\lambda p_0(t)$ with $p_0(0) = 1$, giving $p_0(t) = e^{-\lambda t}$. Hence $J_1 \sim \mathrm{Exp}(\lambda)$.

Q4

Let (X_t) be a Poisson process of rate λ and set $N_t^1 = X_t$, $N_t^2 = X_t$. Both are Poisson(λ), but $N_t^1 + N_t^2 = 2X_t$ takes only even values, whereas any Poisson process of positive rate assigns positive probability to odd integers. Hence the superposition is not a Poisson process.

Q5

Write $\lfloor t \rfloor = n$. By the SLLN applied to the i.i.d. increments $N_k = X_k - X_{k-1} \sim \mathrm{Poi}(\lambda)$,

$$\frac{X_n}{n} = \frac{1}{n}\sum_{k=1}^{n} N_k \xrightarrow{\text{a.s.}} \lambda.$$

Monotonicity of paths gives $X_n \le X_t \le X_{n+1}$ for $n \le t < n+1$, so

$$\frac{X_n}{t} \le \frac{X_t}{t} \le \frac{X_{n+1}}{t}.$$

As $t \to \infty$, $n/t \to 1$ and $(n+1)/t \to 1$, so both bounds converge a.s. to λ by the above. Hence $X_t/t \xrightarrow{\text{a.s.}} \lambda$.

Q6

Since $N^{(1)}$ and $N^{(2)}$ are independent and have no simultaneous jumps a.s., the combined process $(N_t^{(1)}, N_t^{(2)})$ can be viewed as a single Poisson process of rate $\lambda_1 + \lambda_2$ in which each event is independently labelled "type 1" with probability $p = \lambda_1/(\lambda_1+\lambda_2)$ or "type 2" with probability $1-p$. The state (i,j) is visited if and only if among the first $i+j$ combined events exactly i are type 1, giving

$$\mathbb{P}(\text{process visits } (i,j)) = \binom{i+j}{i}\left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^i\left(\frac{\lambda_2}{\lambda_1+\lambda_2}\right)^j.$$

Q7 (Two-salesman CTMC)

States: 0 (nobody on phone), 1 (salesman 1 only), 2 (salesman 2 only), 12 (both).

$$Q = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 12 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 12 \end{array} \\ \begin{pmatrix} -(\lambda_1+\lambda_2) & \lambda_1 & \lambda_2 & 0 \\ \mu_1 & -(\mu_1+\lambda_2) & 0 & \lambda_2 \\ \mu_2 & 0 & -(\mu_2+\lambda_1) & \lambda_1 \\ 0 & \mu_2 & \mu_1 & -(\mu_1+\mu_2) \end{pmatrix} \end{array}.$$

Q8 (Explosion)

Since the chain only jumps from i to $i+1$, the inter-jump times $S_i \sim \mathrm{Exp}(\lambda(i))$ are independent and $\zeta = \sum_{i=0}^{\infty} S_i$. From Practice Set 8 Q2:

(i) $\mathbb{P}(\zeta < \infty) = 1$ if and only if $\displaystyle\sum_{i=0}^{\infty} \frac{1}{\lambda(i)} < \infty$.

(ii) $\mathbb{P}(\zeta = \infty) = 1$ if and only if $\displaystyle\sum_{i=0}^{\infty} \frac{1}{\lambda(i)} = \infty$.

17 Practice Problem Set 10

Q1 (M/M/1/K queue)

(a) Model as a CTMC on $\{0, 1, \ldots, K\}$ with $q_{i,i+1} = \lambda$ for $0 \le i < K$ and $q_{i,i-1} = \mu$ for $1 \le i \le K$.

(b) Let $\rho = \lambda/\mu$. Detailed balance gives $\pi_i = \pi_0 \rho^i$. Normalising,

$$\pi_0 = \frac{1-\rho}{1-\rho^{K+1}} \quad (\rho \neq 1), \qquad \pi_0 = \frac{1}{K+1} \quad (\rho = 1).$$

The fraction of time the server is free is π_0.

Q2 (Machine with two parts)

Let T_1, T_2, T_3 be the times of the first type-1, type-2, type-3 shock respectively, with $T_i \sim \mathrm{Exp}(\lambda_i)$ independent. Part 1 fails at $U = \min(T_1, T_3)$ and part 2 at $V = \min(T_2, T_3)$.

(a) For $s, t \geq 0$,

$$\mathbb{P}(U > s, V > t) = \mathbb{P}(T_1 > s)\mathbb{P}(T_2 > t)\mathbb{P}(T_3 > \max(s,t)) = e^{-\lambda_1 s} e^{-\lambda_2 t} e^{-\lambda_3 \max(s,t)}.$$

(b) Marginals: $U \sim \mathrm{Exp}(\lambda_1 + \lambda_3)$, $V \sim \mathrm{Exp}(\lambda_2 + \lambda_3)$. If U, V were independent, $\mathbb{P}(U > s, V > t)$ would equal $e^{-(\lambda_1+\lambda_3)s} e^{-(\lambda_2+\lambda_3)t}$. Since $e^{-\lambda_3 \max(s,t)} \neq e^{-\lambda_3(s+t)}$ whenever $\lambda_3 > 0$, U and V are *not* independent when $\lambda_3 > 0$.

Q3 (N-component repair system)

(a) From state i (number functioning): failures occur at total rate $i\lambda$, repairs at total rate $(N - i) \cdot 1$.

$$Q_{i,i-1} = i\lambda \ (i \geq 1), \quad Q_{i,i+1} = N - i \ (i \leq N-1), \quad Q_{ii} = -(i\lambda + N - i).$$

(b) Detailed balance $\pi_i \cdot i\lambda = \pi_{i-1}(N - i + 1)$ gives $\pi_i = \pi_0 \binom{N}{i} (1/\lambda)^i$. Normalising,

$$\pi_0 = \left(1 + \frac{1}{\lambda}\right)^{-N}, \quad \pi_i = \binom{N}{i} \left(\frac{1}{1+\lambda}\right)^i \left(\frac{\lambda}{1+\lambda}\right)^{N-i}.$$

So $X_t \sim \mathrm{Bin}\left(N, \frac{1}{1+\lambda}\right)$ in stationarity.

Q4

The Markov property of (X_t) gives independence of X_{t+h} from $(X_s, s \leq t)$ given X_t. By the semigroup $P(h) = e^{Qh} = I + Qh + O(h^2)$,

$$\mathbb{P}(X_{t+h} = j \mid X_t = i) = \delta_{i,j} + q_{i,j} h + o(h) \quad \text{uniformly in } t.$$

Q5

(a) If $i \to j$ in (X_t) there is a sequence of states and positive-rate jumps leading from i to j; these same jumps correspond to transitions in $\{Y_n\}$, so $i \to j$ in $\{Y_n\}$.

(b) If i is recurrent for $\{Y_n\}$, the embedded chain returns to i infinitely often. Each return cycle contributes at least the holding time at i, which is $\mathrm{Exp}(q_i) > 0$. Hence $J_n \geq \sum_{\text{returns}} H_i \to \infty$ a.s., so $\zeta = \infty$ a.s.

Q6

(a) State i is recurrent iff $q_i = 0$ (absorbing, trivially returns) or $\mathbb{P}_i(T_i < \infty) = 1$ (a.s. return after leaving).

(b) State i is transient iff $q_i > 0$ and $\mathbb{P}_i(T_i < \infty) < 1$.

Q7 (Pos. recurrence discrepancy)

(a) *CTMC positive recurrent, embedded DTMC null recurrent.* Take $\mathbb{Z}_+$ with $q_{i,i+1} = q_{i,i-1} = i^3$ for $i \geq 1$ and $q_{0,1} = 1$. The embedded DTMC is the symmetric nearest-neighbour walk on $\mathbb{Z}_+$ (reflected at 0), which is null recurrent. In continuous time, the holding time at state i is $\mathrm{Exp}(2i^3)$, so $\mathbb{E}_1[T_0] \leq 1/i^2 \to 0$, giving $\mathbb{E}_0[T_0^+] < \infty$ and positive recurrence.

(b) *Embedded DTMC positive recurrent, CTMC null recurrent.* Take $\mathbb{Z}_+$ with $q_{0,i} = 1/i^2$ (for $i \geq 1$, so $q_0 = \pi^2/6$) and $q_{i,0} = 1/i^2$ for $i \geq 1$. The embedded DTMC returns to 0 in exactly 2 steps (pos. recurrent). The CTMC mean return time is $\mathbb{E}_0[T_0^+] = 1/q_0 + \sum_{i\geq 1} p_{0i}/q_i \geq \sum_{i\geq 1}(6/\pi^2 i^2) \cdot i^2 = \infty$.

Q8 (M/M/1 positive recurrence)

The M/M/1 queue is an irreducible birth–death chain on $\mathbb{Z}_+$. The stationary equation $\pi Q = 0$ and detailed balance give $\pi_i = \rho^i \pi_0$ where $\rho = \lambda/\mu$. Normalisation $\sum_{i\geq 0} \pi_i = 1$ holds iff $\rho < 1$, i.e. $\lambda < \mu$. When $\rho < 1$, $\pi_0 = 1 - \rho > 0$ and the chain is positive recurrent. When $\rho \geq 1$, no probability distribution satisfies $\pi Q = 0$ with $\sum \pi_i = 1$, so the chain is null recurrent or transient.

Q9

Irreducibility of Q implies $P_t(i,j) > 0$ for all $t > 0$ and all i, j; in particular $P_{nh}(i,j) > 0$ for some n, so $\{Z_n\}$ is irreducible. Aperiodicity: $P_h(i,i) = e^{-q_i h} + \cdots > 0$ for all i and $h > 0$, giving $p_{ii}^{(1)} > 0$ in $\{Z_n\}$, hence period 1.

Q10 (Impatient customers)

(a) From state $n \geq 2$: arrivals at rate λ ($n \to n+1$); service completion at rate μ ($n \to n-1$); each of $n-1$ waiting customers abandons at rate δ, total $\to n-1$ at rate $\mu + (n-1)\delta$. From state 1: $1 \to 0$ at rate μ; $1 \to 2$ at rate λ. From 0: $0 \to 1$ at rate λ.

(b) For $\delta = \mu$: the rate from n to $n-1$ becomes $\mu + (n-1)\mu = n\mu$. Detailed balance $\lambda \pi_n = n\mu\, \pi_{n+1}$ gives $\pi_{n+1} = (\lambda/(n+1)\mu)\pi_n = (\rho/(n+1))\pi_n$. By induction $\pi_n = \pi_0 \rho^n / n!$. Normalising, $\pi_0 e^\rho = 1$, so

$$\pi_n = e^{-\rho} \frac{\rho^n}{n!}, \quad \rho = \lambda/\mu.$$

The stationary distribution is Poisson(ρ).

Q11

The chain is irreducible (path $0 \to 1 \to 2 \to 0$ exists with positive rates), so the limit equals the stationary probability π_2. Solving $\pi Q = 0$:

$$-3\pi_0 + 2\pi_2 = 0 \implies \pi_2 = \tfrac{3}{2}\pi_0; \quad \pi_0 - \pi_1 = 0 \implies \pi_1 = \pi_0.$$

Normalisation: $\pi_0 + \pi_0 + \frac{3}{2}\pi_0 = \frac{7}{2}\pi_0 = 1$, so $\pi_0 = 2/7$ and

$$\lim_{t\to\infty} \mathbb{P}(X_t = 2 \mid X_0 = 1) = \pi_2 = \frac{3}{7}.$$

18 Practice Problem Set 11

Q1

Reading assignment: Example 3.5.4 in Norris.

Q2 (Output of stationary M/M/1)

Let $\rho = \lambda/\mu < 1$. The departure rate at time t is $\mu \cdot \mathbf{1}_{\{X_t \geq 1\}}$. Under stationarity, $\mathbb{E}[\mu \mathbf{1}_{\{X_t \geq 1\}}] = \mu(1 - \pi_0) = \mu\rho = \lambda$.

To show the departure process $\{Y_t\}$ is Poisson(λ) without reversibility: use the Markov property and stationarity to show the increments $Y_{t_2} - Y_{t_1}$ are independent across disjoint intervals. Given X_s starts from π at time s, the future evolution is stationary, so each inter-departure interval has the same distribution. A departure epoch restarts the queue in state $X_{\tau-} - 1$ which, under stationarity, has distribution $\pi_{k-1} \propto (1-\rho)\rho^{k-1}$. A careful computation via the Kolmogorov forward equations shows the counting process has Laplace functional equal to that of a Poisson(λ) process; the key is that the stationary rate is λ and the process has no memory.

Q3 ($M/M/\infty$ queue)

(a) Each of the n occupied lines completes at rate μ, so $q_{n,n-1} = n\mu$; arrivals give $q_{n,n+1} = \lambda$.

(b) Detailed balance $\pi_n \cdot n\mu = \pi_{n-1} \cdot \lambda$ gives $\pi_n = (\lambda/\mu)^n/(n!) \cdot \pi_0$. Normalisation: $\pi_0 e^{\lambda/\mu} = 1$, so

$$\pi_n = e^{-\lambda/\mu} \frac{(\lambda/\mu)^n}{n!} \sim \text{Poisson}(\lambda/\mu).$$

(c) By Burke's theorem (or time-reversal), the departure process in stationarity is Poisson(λ), so $Y_t \sim \text{Poisson}(\lambda t)$.

Q4 (Restricted CTMC)

For $x, y \in A$, detailed balance for (X_t) gives $\pi(x)q_{x,y} = \pi(y)q_{y,x}$. Since $\bar{q}_{x,y} = q_{x,y}$ for $x, y \in A$,

$$\nu(x)\bar{q}_{x,y} = \frac{\pi(x)}{C} q_{x,y} = \frac{\pi(y)}{C} q_{y,x} = \nu(y)\bar{q}_{y,x}.$$

So ν satisfies detailed balance for $\{Y_t\}$. Since $\sum_{x\in A} \nu(x) = 1$, ν is the stationary distribution.

Q5 (Tandem queues)

(a) By Burke's theorem, the output of a stable M/M/1 queue is Poisson at the input rate. The output of queue 1 is Poisson(λ); thinning it by p gives arrivals to queue 2 as Poisson($p\lambda$).

(b) Stability requires $\lambda < \mu_1$ and $p\lambda < \mu_2$. Let $\rho_1 = \lambda/\mu_1$, $\rho_2 = p\lambda/\mu_2$. By Jackson's theorem (or the product-form result for tandem queues with independent routing),

$$\pi(n_1, n_2) = (1-\rho_1)\rho_1^{n_1}(1-\rho_2)\rho_2^{n_2}, \quad n_1, n_2 \geq 0.$$

Q6 (Jackson network)

(a) Write the traffic equations as $(I - P^\top)\beta = \lambda$ where P_{ij} is the routing probability from j to i. Since the network is open (all customers eventually exit), the spectral radius of P is strictly less than 1, so $(I - P^\top)$ is invertible with non-negative inverse. Hence $\beta = (I - P^\top)^{-1}\lambda$ is the unique non-negative solution.

(b) If $\rho_i = \beta_i/\mu_i < 1$ for all i, one verifies that $\pi(\mathbf{n}) = \prod_{i=1}^{K}(1-\rho_i)\rho_i^{n_i}$ satisfies the global balance equations $\pi Q = 0$, using the fact that the routing structure decouples the queues in stationarity.

Q7 (Conditional expectation from joint density)

$Y(\omega)$ is $\sigma(Z)$-measurable since it is a function of $Z(\omega)$. For any $B = \{Z \in C\} \in \sigma(Z)$,

$$\mathbb{E}[Y\mathbf{1}_B] = \int_C \frac{\int g(x)f(x,z)\,dx}{\int f(x,z)\,dx} \int f(x,z)\,dx\,dz = \int_C \int g(x)f(x,z)\,dx\,dz = \mathbb{E}[g(X)\mathbf{1}_B].$$

Hence Y is a version of $\mathbb{E}[g(X) \mid Z]$.

Q8

For any $A \in \mathcal{G}$ with $A \subseteq B$,

$$\mathbb{E}[\mathbb{E}[X_1 \mid \mathcal{G}]\mathbf{1}_A] = \mathbb{E}[X_1\mathbf{1}_A] = \mathbb{E}[X_2\mathbf{1}_A] = \mathbb{E}[\mathbb{E}[X_2 \mid \mathcal{G}]\mathbf{1}_A].$$

Since $A \in \mathcal{G}$, $A \subseteq B$ are generating sets for $\mathcal{G}$ restricted to B, the a.s. equality $\mathbb{E}[X_1 \mid \mathcal{G}] = \mathbb{E}[X_2 \mid \mathcal{G}]$ holds on B.

Q9

(a) **Linearity.** For any $A \in \mathcal{G}$,

$$\mathbb{E}[(\alpha\mathbb{E}[X_1 \mid \mathcal{G}] + \beta\mathbb{E}[X_2 \mid \mathcal{G}])\mathbf{1}_A] = \alpha\mathbb{E}[X_1\mathbf{1}_A] + \beta\mathbb{E}[X_2\mathbf{1}_A] = \mathbb{E}[(\alpha X_1 + \beta X_2)\mathbf{1}_A].$$

The left side is $\mathcal{G}$-measurable, confirming it is the conditional expectation.

(b) **Positivity.** Let $D = X_2 - X_1 \geq 0$. For any $A \in \mathcal{G}$, $\mathbb{E}[\mathbb{E}[D \mid \mathcal{G}]\mathbf{1}_A] = \mathbb{E}[D\mathbf{1}_A] \geq 0$. A non-negative $\mathcal{G}$-measurable function integrating to ≥ 0 over every $A \in \mathcal{G}$ must be a.s. non-negative. Hence $\mathbb{E}[X_1 \mid \mathcal{G}] \leq \mathbb{E}[X_2 \mid \mathcal{G}]$ a.s.

Q10 (Wald's equations)

(a) Conditioning on N: $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid N]] = \mathbb{E}[N\mathbb{E}[Y_1]] = \mu\mathbb{E}[N]$.

(b) $\mathbb{E}[X^2] = \mathbb{E}[\mathbb{E}[(\sum_{i=1}^{N} Y_i)^2 \mid N]]$. Given $N = n$, the sum has variance $n\sigma^2$ and mean $n\mu$, so $\mathbb{E}[(\sum Y_i)^2 \mid N] = N\sigma^2 + N^2\mu^2$. Thus

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \sigma^2\mathbb{E}[N] + \mu^2\mathbb{E}[N^2] - \mu^2(\mathbb{E}[N])^2 = \sigma^2\mathbb{E}[N] + \mu^2\mathrm{Var}(N).$$

19 Practice Problem Set 12

Q1 (Conditional expectation on partition)

Define $Y = \sum_{i\geq 1} \frac{\mathbb{E}[X\mathbf{1}_{B_i}]}{\mathbb{P}(B_i)}\mathbf{1}_{B_i}$. Clearly Y is $\mathcal{G}$-measurable. For any $A \in \mathcal{G}$ write $A = \bigcup_{i\in I} B_i$ (disjoint union for some index set I):

$$\mathbb{E}[Y\mathbf{1}_A] = \sum_{i\in I} \frac{\mathbb{E}[X\mathbf{1}_{B_i}]}{\mathbb{P}(B_i)}\mathbb{P}(B_i) = \sum_{i\in I} \mathbb{E}[X\mathbf{1}_{B_i}] = \mathbb{E}[X\mathbf{1}_A].$$

Hence $Y = \mathbb{E}[X \mid \mathcal{G}]$, and $Y = \mathbb{E}[X\mathbf{1}_{B_i}]/\mathbb{P}(B_i)$ on B_i.

Q2

Since $\mathcal{G}_n \subseteq \mathcal{F}_n$ and $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n$,

$$\mathbb{E}[X_{n+1} \mid \mathcal{G}_n] = \mathbb{E}[\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \mid \mathcal{G}_n] = \mathbb{E}[X_n \mid \mathcal{G}_n] = X_n,$$

since X_n is $\mathcal{G}_n$-measurable. So $\{X_n\}$ is a martingale w.r.t. $\{\mathcal{G}_n\}$.

Q3 (Generator martingale)

$$\begin{aligned}
\mathbb{E}[Y_{n+1} \mid \sigma(X_0, \ldots, X_n)] &= \mathbb{E}[f(X_{n+1}) \mid X_n] - f(X_0) - \sum_{i=0}^{n-1} (\mathcal{L}f)(X_i) - (\mathcal{L}f)(X_n) \\
&= \sum_y p_{X_n,y} f(y) - f(X_0) - \sum_{i=0}^{n-1} (\mathcal{L}f)(X_i) - (\mathcal{L}f)(X_n) \\
&= \big[f(X_n) + (\mathcal{L}f)(X_n)\big] - f(X_0) - \sum_{i=0}^{n-1} (\mathcal{L}f)(X_i) - (\mathcal{L}f)(X_n) = Y_n.
\end{aligned}$$

Q4 (Galton–Watson martingale)

$$\mathbb{E}\Big[\mu^{-(n+1)} X_{n+1} \mid \sigma(Z^1, \ldots, Z^n)\Big] = \mu^{-(n+1)}\mathbb{E}[X_{n+1} \mid X_n] = \mu^{-(n+1)} \cdot \mu X_n = \mu^{-n} X_n.$$

Q5 (Superharmonic function)

$$\mathbb{E}[X_{n+1} \mid S_1, \ldots, S_n] = \mathbb{E}[f(S_n + Z_{n+1}) \mid S_n] = \frac{1}{\mathrm{vol}(B(0,1))} \int_{B(S_n,1)} f(y)\, dy \leq f(S_n) = X_n,$$

where the inequality is superharmonicity. Hence $\{X_n\}$ is a supermartingale.

Q6

$$\mathbb{E}[\lambda^{-(n+1)} f(X_{n+1}) \mid X_0, \ldots, X_n] = \lambda^{-(n+1)} \sum_y p_{X_n,y} f(y) \leq \lambda^{-(n+1)} \cdot \lambda f(X_n) = \lambda^{-n} f(X_n).$$

Hence $\{\lambda^{-n} f(X_n)\}$ is a supermartingale.

Q7 (Pólya's urn)

(a) $\mathbb{E}[Y_{n+1} \mid R_1, \ldots, R_n] = \mathbb{E}\left[\frac{R_{n+1}}{n+1+r+b} \,\middle|\, R_n\right] = \frac{R_n \cdot (n+r+b+1)/(n+r+b)}{n+r+b+1} = \frac{R_n}{n+r+b} = Y_n$.

(b) With $r = b = 1$: $Y_n = R_n/(n+2)$ is a bounded martingale and $T < \infty$ a.s. (since $\sum P(T = k) = \sum 1/(k(k+1)) = 1$). At time T (first blue draw), $R_T = T$ (one started with 1 red and $T-1$ red balls were added), so $Y_T = T/(T+2)$. By the optional stopping theorem,

$$\mathbb{E}[Y_T] = Y_0 = \frac{1}{2} \implies \mathbb{E}\left[\frac{T}{T+2}\right] = \frac{1}{2} \implies 1 - \mathbb{E}\left[\frac{2}{T+2}\right] = \frac{1}{2} \implies \mathbb{E}\big[(T+2)^{-1}\big] = \frac{1}{4}.$$

Q8 (Symmetric random walk hitting time)

(a) Since $\{X_n\}$ is a martingale and $T < \infty$ a.s. (bounded stopping time in the limit), OST gives $\mathbb{E}[X_T] = 0$. With $p = \mathbb{P}(X_T = -a)$,

$$b(1-p) + (-a)p = 0 \implies p = \frac{b}{a+b}.$$

(b) $\{X_n^2 - n\}$ is a martingale. OST gives $\mathbb{E}[X_T^2] = \mathbb{E}[T]$. Now

$$\mathbb{E}[X_T^2] = b^2 \cdot \frac{a}{a+b} + a^2 \cdot \frac{b}{a+b} = \frac{ab(a+b)}{a+b} = ab.$$

Hence $\mathbb{E}[T] = ab$.

20 Practice Problem Set 13

P1 (Hitting probability is harmonic / martingale)

Define $g(i) = \mathbb{P}_i(\tau_N^+ < \infty)$ with $\tau_N^+ = \inf\{m \geq 1 : X_m = N\}$. First-step analysis gives

$$g(i) = \sum_j p_{ij} g(j) \quad \text{for every state } i.$$

Hence

$$\mathbb{E}[g(X_{n+1}) \mid \mathcal{F}_n] = \sum_j p_{X_n,j} g(j) = g(X_n),$$

so $\{g(X_n)\}$ is a martingale w.r.t. $\{\mathcal{F}_n\}$.

P2 (Wald's lemma counterexample)

Let X_n be i.i.d. Rademacher: $\mathbb{P}(X_n = \pm 1) = \frac{1}{2}$, so $\mathbb{E}[X_1] = 0$. Define

$$N = \begin{cases} 1, & X_2 = 1, \\ 2, & X_2 = -1. \end{cases}$$

Then $\mathbb{E}[N] = \frac{3}{2}$ and $\mathbb{E}[N]\mathbb{E}[X_1] = 0$. But

$$\mathbb{E}\left[\sum_{n=1}^{N} X_n\right] = \mathbb{E}[X_1] + \mathbb{E}[\mathbf{1}_{\{X_2=-1\}} X_2] = 0 + (-1) \cdot \tfrac{1}{2} = -\tfrac{1}{2} \neq 0.$$

Wald's lemma fails because N depends on the future value X_2, so N is not a stopping time.

P3 (Convex inequality)

For any $x \in [-a, b]$, write $x = \frac{b-x}{a+b}(-a) + \frac{a+x}{a+b}b$ as a convex combination. Convexity of f gives $f(x) \le \frac{b-x}{a+b}f(-a) + \frac{a+x}{a+b}f(b)$. Taking expectations and using $\mathbb{E}[X] = 0$:

$$\mathbb{E}[f(X)] \le \frac{b - \mathbb{E}[X]}{a+b}f(-a) + \frac{a + \mathbb{E}[X]}{a+b}f(b) = \frac{bf(-a) + af(b)}{a+b}.$$

P4 (Critical Galton–Watson dies out)

Let $\phi(s) = \mathbb{E}[s^Z]$ be the offspring pgf with $\phi'(1) = \mu = 1$. The extinction probability q is the smallest fixed point of ϕ in $[0, 1]$. Define $h(s) = \phi(s) - s$; then $h(1) = 0$ and $h'(1) = 0$. Since $\mathbb{P}(Z = 1) < 1$ and $\mathbb{E}[Z] = 1$, there must be offspring mass at $k \ge 2$, giving $\phi''(s) = \mathbb{E}[Z(Z-1)s^{Z-2}] > 0$ for $s \in (0, 1)$. So ϕ is strictly convex, and $h(s) > 0$ for all $s \in [0, 1)$. Hence $\phi(s) > s$ for $s < 1$, meaning $q = 1$.

P5 (Concentration for shortest path)

The points lie on the unit circle, so arranging them cyclically and connecting consecutive arcs gives a Hamiltonian cycle of length $\le 2\pi$. Deleting the longest edge gives a path through all n points of length $\le 2\pi$. Hence $0 \le T \le 2\pi < 8$. Applying Hoeffding's inequality with range $[0, 2\pi]$:

$$\mathbb{P}(|T - \mathbb{E}[T]| \ge a) \le 2\exp\left(-\frac{2a^2}{(2\pi)^2}\right) \le 2\exp\left(-\frac{a^2}{32}\right).$$

P6 (McDiarmid for Hamming distance)

Define $f(x_1, \ldots, x_n) = \min_{y \in A} d_H(x, y)$. Changing one coordinate of x changes $d_H(x, y)$ by at most 1 for every fixed y, so $|f(x) - f(x')| \le 1$ whenever x, x' differ in one coordinate. By McDiarmid's inequality with $c_i = 1$:

$$\mathbb{P}(D \ge a) = \mathbb{P}(D - \mu \ge a - \mu) \le \exp\left(-\frac{2(a-\mu)^2}{n}\right), \quad a > \mu.$$

P7 (Poisson branching process)

Given $X_{n-1} = k$, $X_n \sim \text{Poisson}(k) = \sum_{i=1}^{k} Y_i$ with $Y_i \overset{\text{i.i.d.}}{\sim} \text{Poisson}(1)$. So $\{X_n\}$ is a Galton–Watson process with offspring $Z \sim \text{Poisson}(1)$: mean $\mu = 1$, $\mathbb{P}(Z = 1) = e^{-1} < 1$. By P4, the extinction probability is $q = 1$, so $X_n \to 0$ a.s. In fact $\mathbb{P}(X_n = 0) \to 1$. Note that $\mathbb{E}[X_n] = 1$ for all n (the process is a martingale), illustrating that a.s. extinction coexists with constant mean.

21 Quiz 8

P1 (True/False)

1. **False.** X_s and X_t are not independent: $X_t = X_s + (X_t - X_s)$ implies $X_t \ge X_s$, so $\mathbb{P}(X_t < X_s) = 0$ yet $\mathbb{P}(X_t < 5) > 0$ unconditionally when $X_s = 5$.

2. **False.** $X_t - X_s$ and X_{t-s} are identically distributed (both Poisson($\lambda(t-s)$) by stationary increments) but are *not the same random variable*: $X_t - X_s$ counts arrivals in $(s, t]$ while X_{t-s} counts arrivals in $(0, t-s]$.

P2 (Finite-dimensional distributions)

For $0 = t_0 < t_1 < \cdots < t_n$ and $0 \le x_1 \le \cdots \le x_n$ (integers):

$$\mathbb{P}(X_{t_1} = x_1, \ldots, X_{t_n} = x_n) = \prod_{i=1}^{n} \mathbb{P}(X_{t_i} - X_{t_{i-1}} = x_i - x_{i-1}) = e^{-\lambda t_n} \prod_{i=1}^{n} \frac{(\lambda(t_i - t_{i-1}))^{x_i - x_{i-1}}}{(x_i - x_{i-1})!}.$$

This closed-form formula in terms of λ, t_i, and x_i alone shows that all finite-dimensional distributions of a Poisson(λ) process are completely determined.

22 Quiz 9

P1

For fixed $t > 0$, write $X_{nt} = \sum_{k=1}^{n}(X_{kt} - X_{(k-1)t})$. The increments are i.i.d. Poisson(λt) with mean λt. By the SLLN,

$$\frac{X_{nt}}{n} = \frac{1}{n}\sum_{k=1}^{n}(X_{kt} - X_{(k-1)t}) \xrightarrow{\text{a.s.}} \lambda t.$$

P2 (True/False)

1. **False.** The even arrivals form a *renewal process* but not a Poisson process. The inter-event times for the even-arrival subsequence are sums of two independent Exp(λ) variables (Gamma$(2, \lambda)$), which are not exponential.

2. **False.** The locations of arrivals in $[s, t]$ are *not* uniformly distributed on $[s, t]$. Conditioned on the number of arrivals in $[s, t]$, the locations are uniform on $(s, t]$, but the unconditional statement is false: there are no arrivals in $[s, t]$ with probability $e^{-\lambda(t-s)} > 0$.

P3 (Fill in the blanks)

1. The holding time in state i has exponential distribution with parameter $\mathbf{q_i} = |\mathbf{q_{ii}}|$ (the total rate out of state i).

2. In the embedded chain of an M/M/1 queue, from state 1 the chain jumps to 2 (arrival before service) with probability $\lambda/(\lambda + \mu)$.

23 Quiz 10

P1

The CTMC with $Q = [(-3, 2, 1); (0, -1, 1); (1, 0, -1)]$ is irreducible (path $0 \to 1 \to 2 \to 0$), so $\lim_{t\to\infty} \mathbb{P}(X_t = 2 \mid X_0 = 1) = \pi_2$. Solving $\pi Q = 0$:

$$-3\pi_0 + \pi_2 = 0 \implies \pi_2 = 3\pi_0; \quad 2\pi_0 - \pi_1 = 0 \implies \pi_1 = 2\pi_0.$$

Normalisation: $6\pi_0 = 1$, giving $\pi_0 = 1/6$ and $\pi_2 = 1/2$.

P2 (True/False)

1. **True.** For a finite irreducible CTMC, the stationary distribution π has $\pi_i > 0$ for all i. The mean return time in continuous time is $1/(\pi_i v_i) < \infty$ (where $v_i = q_i$), and the embedded DTMC has stationary distribution $y_i \propto \pi_i v_i > 0$, so it is also positive recurrent. The two notions are equivalent for finite irreducible chains.

2. **False.** A time-sampled process $Z_n = X_{T_n}$ need not be Markov if (T_n) are not fixed multiples of h but instead depend on the path. Counterexample: let T_3 depend on Z_1; then $P(Z_3 = j \mid Z_2 = i, Z_1 = k)$ need not equal $P(Z_3 = j \mid Z_2 = i)$.

24 Quiz 11

P1 (Positivity of conditional expectation)

Let $D = X_2 - X_1 \geq 0$. By linearity, $\mathbb{E}[X_2 \mid \mathcal{G}] - \mathbb{E}[X_1 \mid \mathcal{G}] = \mathbb{E}[D \mid \mathcal{G}]$. For any $A \in \mathcal{G}$,

$$\mathbb{E}[\mathbb{E}[D \mid \mathcal{G}]\mathbf{1}_A] = \mathbb{E}[D\mathbf{1}_A] \geq 0.$$

A $\mathcal{G}$-measurable random variable whose integral over every $A \in \mathcal{G}$ is ≥ 0 must be a.s. non-negative. Hence $\mathbb{E}[X_1 \mid \mathcal{G}] \leq \mathbb{E}[X_2 \mid \mathcal{G}]$ a.s.

P2

1. **Two-state CTMC is reversible.** Let the states be $\{1, 2\}$ with rates $q_{12} = \alpha$, $q_{21} = \beta$. The stationary distribution satisfies $\pi_1\alpha = \pi_2\beta$ (flow balance), i.e. $\pi_1 q_{12} = \pi_2 q_{21}$. This is exactly the detailed balance condition, so the chain is reversible w.r.t. π.

2. **Example of irreducible non-reversible CTMC.** Take states $\{1, 2, 3\}$ with $q_{12} = q_{23} = q_{31} = 1$ and all other off-diagonal rates 0. The unique stationary distribution is uniform ($\pi_i = 1/3$). The Kolmogorov cycle condition requires $q_{12}q_{23}q_{31} = q_{13}q_{32}q_{21}$; the left side equals 1, the right side equals 0. Hence the chain is not reversible.

25 Quiz 12

P1

Use the martingale $\{X_n\}$ (symmetric random walk is a martingale). Since $T < \infty$ a.s. and $\{X_{T\wedge n}\}$ is bounded in L^1, OST gives $\mathbb{E}[X_T] = X_0 = 0$. Let $p = \mathbb{P}(X_T = -a)$:

$$b(1-p) + (-a)p = 0 \implies \mathbb{P}(X_T = -a) = \frac{b}{a+b}.$$

P2 (True/False)

1. **True.** For any $A \in \mathcal{F}_n$, $\mathbb{E}[(X_{n+1} + Y_{n+1}) \mid \mathcal{F}_n] = \mathbb{E}[X_{n+1} \mid \mathcal{F}_n] + \mathbb{E}[Y_{n+1} \mid \mathcal{F}_n] \geq X_n + Y_n$, so $\{X_n + Y_n\}$ is a submartingale.

2. **False** in general. The correct statement requires $H_n \geq 0$. Counterexample: $X_n = n$ (deterministic submartingale) and $H_n = -1$ (predictable, negative). Then $(H \cdot X)_n = -n$ satisfies $\mathbb{E}[(H \cdot X)_{n+1} \mid \mathcal{F}_n] = -(n+1) < -n = (H \cdot X)_n$, so it is a *supermartingale*, not a submartingale.

26 Quiz 13

P1

See Practice Problem Set 13, P7. The sequence $\{X_n\}$ is a Galton–Watson branching process with offspring distribution Poisson(1) (critical, non-degenerate). Extinction probability $q = 1$, so $X_n \to 0$ a.s. and $\mathbb{P}(X_n = 0) \to 1$. The mean remains $\mathbb{E}[X_n] = 1$ for all n.

P2 (True/False)

1. **False.** The correct statement requires φ to be convex *and non-decreasing*. Counterexample: $X_n = n$ (deterministic submartingale), $\varphi(x) = (x-3)^2$ (convex but not monotone). Then $\varphi(X_1) = 4$ but $\varphi(X_2) = 1$, so $\mathbb{E}[\varphi(X_2) \mid \mathcal{F}_1] = 1 < 4 = \varphi(X_1)$: not a submartingale.

2. **False.** Uniform integrability is needed for L^1 convergence. Counterexample: Let $X_n = 2^n \mathbf{1}_{U \le 2^{-n}}$ where $U \sim \mathrm{Unif}(0,1)$. Then $\{X_n\}$ is a martingale with $\mathbb{E}[|X_n|] = 1$ for all n, so $\sup_n \mathbb{E}[|X_n|] = 1 < \infty$. But $X_n \to 0$ a.s. while $\mathbb{E}[|X_n - 0|] = 1 \not\to 0$, so X_n does not converge in L^1.